Rakshitha B Y

No. 813, Shree Vishnu Nilaya, 11th Cross, D Group Layout, Andrahalli Main Road,
Lingadheeranahalli Bengaluru – 560091, Karnataka, India
Ph: +91 8296700346; E-mail: rakshh.29@gmail.com

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ACADEMIC QUALIFICATION

Bachelor of Engineering in **Information Science**,
J.S.S. Academy of Technical Education, Bangalore;
Aggregate/CGPA: 8.5(ongoing)
Duration: 2022 – 2026

INTERNSHIP

Software Developer Intern, InternPe, Bengaluru ***Feb 2025 – Mar 2025***

- Completed internship in Java Programming.
- Recognized for being genuine, committed, and demonstrating excellent job skills.

ACADEMIC PROJECTS

Title: Smart Health Monitoring Belt for Immobile Patients
Duration: Feb 2025 – Present
Team size: 4
Summary: Creating a novel wearable device for monitoring vital signs (heartbeat, body temperature and motion) in hemiplegic and cardio patients. Integrated pulse, temperature, and accelerometer sensors with a GSM module. Enabled these patients to continuously monitor vitals in real time (IoT platform).
Individual Contribution:

- Conceptualization: Defined project concepts, scope, and objectives.
- Methodology: Defined a development strategy, tools, and logic of work.
- Data Curation: Developed logic for tracking income/expense organizationally.
- Resources: Developed and gathered visual and technical GPU assets.
- Writing - Review & editing: Drafted and edited project documents and reports.

Title: AI-Enhanced Kernel for Dynamic Resource Management
Duration: Sep 2024 – Jan 2025
Team size: 4
Summary: Created an AI-enhanced Linux kernel for real-time resource distribution in virtual environments. Used a random forest model to forecast and optimize CPU and memory usage dynamically, minimizing resource waste and boosting system stability. Incorporated control groups (cgroups) and the psutil library for live system monitoring. Reached 99.34% accuracy in resource prediction, showing notable gains in efficiency.
Individual Contribution:

- Software Development: Implemented system resource tracking modules using Python (psutil).
- Validation: Verified prediction results against live system metrics.
- Writing – Review & Editing: Contributed to reviewing and refining the research manuscript for submission.

INDEPENDENT PROJECT

Title: Expense Tracker Dashboard
Duration: Mar 2025 – Apr 2025
Individual Project
Summary: Creating an expense tracking online interface that is visually appealing and responsive in design with a fake MERN backend. The real time income and expense tracking, along with the ability to manipulate the DOM in real time and assess input validity for client-side errors are some features of the

project. Build primarily using HTML5, CSS3, and JavaScript (ES6), with a fake indication of a server interaction within the user interface that returns a "Hello from MERN" string. In addition to developing a pastel-themed user interface, the project demonstrates applicant's thinking of the application logic for curation and display of data. This is an ongoing project; however, it establishes the applicant's attitudes to strong front-end development practices, interest in aesthetic design elements, and foundational level completion of MERN.

PUBLICATIONS

- *Co-author, "AI-Augmented Kernel for Dynamic Resource Utilization in Virtualized Environments", submitted to* ***Engineering, Technology & Applied Science Research (ETSAR)****, Submission ID:* ***#12536****, June 2025.*

AWARDS & ACHIEVEMENTS

- Participated in **Admaya Chetana's Green Sunday**, achieving 487 continuous tree planting - Apr 2025.
- Participated in the **School Bell Campaign** by Yuvaka Sangha at GHPS, Gaanakal as part of my commitment to education - May 2025.
- Presented project " **AI-Enhanced Kernel for Dynamic Resource Management** " at **Anveshana 2025: Project Expo hosted by IEEE & ITS Bangalore Chapter - 2025**.
- Awarded with appreciation and a certificate for being **an NSS Volunteer** and active participation in social service.

SEMINARS & CONFERENCES

- **Project Presentation** at **Anveshana 2025 – Intercollegiate Project Expo**, organized by IEEE & ITS Bangalore Chapter, Bengaluru.

TECHNICAL SKILLS

- **Languages**: Java, Python, C, JavaScript, HTML, CSS
- **Web Technologies**: HTML5, CSS3, JavaScript (ES6), DOM Manipulation, MERN (Simulated)
- **Database**: ThingSpeak (IoT), Local Storage (Frontend-based)
- **Developer Tools**: Visual Studio Code, Git, GitHub, Chrome DevTools, Live Server
- **Others**: Data Structures & Algorithms, IoT Integration (GSM, Sensors), Machine Learning (Random Forest), psutil, cgroups, UI/UX Design, Technical Writing

EXTRACURRICULAR ACTIVITIES

- **NSS Volunteer**, JSSATE-Bangalore **(Aug 2023 – Present)** - Applications of social service and outreach initiatives through the National Service Scheme (NSS), with a key focus on sustainability and community welfare.
    - Participated in **tree plantation events**: 487 ongoing Green Sundays organized by Adamya Chetana **(April, 2025)**
    - Volunteered for the **School Bell Campaign** to endorse the learning environments of government schools **(May, 2025)**
    - Participated in several **Clean Sunday events** and in several **walkathon events** relating to the principles of cleanliness and fitness around the community
    - Completed **3K Marathon** at college to promote health and wellness
    - Assisted in organizing **awareness drive** with such issues around environmental and social responsibility on, and off campus

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Mention significant activities along with their dates, locations, durations.
Specify your contribution to/performance in them. Mention achievements in them, if any.
This IS NOT hobby section